

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY, LONERE**

Winter Semester Examination: Dec.-2019

B. Tech. Second Year

Mechanics of Solids (BTCVC 302)

Date: 12/12/2019

Sem.: III

Marks: 60

Time: 3 Hrs

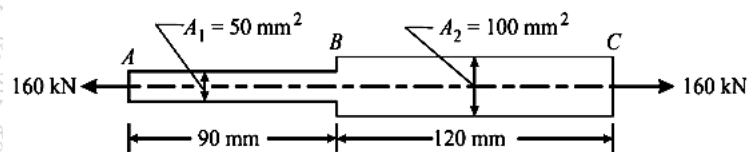
INSTRUCTIONS TO THE STUDENTS

1. Attempt any five questions from the given six questions. Figures to the right indicate full marks.
2. Illustrate your answers with neat sketches, diagrams etc. wherever necessary.
3. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that data is a part of the examination.
4. If some part or parameter is noticed to be missing, you may appropriately assume and mention it.
5. Use of logarithmic table, drawing instruments and non programmable calculators is permitted.

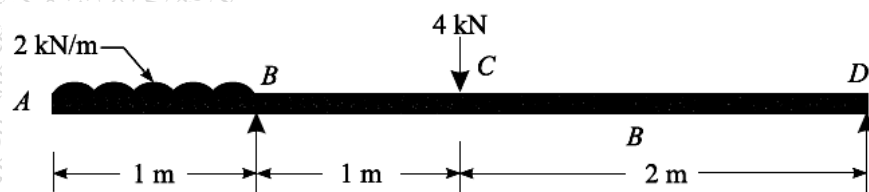
(Marks)

Q. 1 Answer any two from the following:

- a) Define: Hooke's law, Elasticity, and Ductility. (6)
- b) A steel rod 25 mm diameter is at a temperature of 100°C . At this temperature the length of rod is 6 m and is fixed at both of its ends. Determine the stress in the rod, when the temperature falls down to 60°C , if
(a) the ends do not yield, and (b) the ends yield by 1 mm
Take $E = 200 \text{ GPa}$ and $\alpha = 12 \times 10^{-6}$ per degree Celsius (6)
- c) A steel rod ABC is 210 mm long is subjected to 160 kN force as shown in figure. If the value of Young's modulus for the steel is 200 GPa, determine its deformation. (6)



- Q. 2**
- a) Explain the variation of shear force diagram and bending moment diagram at a point or between any two sections of a beam with respect to type of loading? (4)
 - b) A beam 4 m long is overhanging by 1 m and carries load as shown in figure. Calculate the shear force and bending moment at important locations and draw the SFD and BMD for the beam. Also, locate the point of contraflexure. (8)



- Q. 3**
- a) A solid steel shaft has to transmit 100 kW at 160 r.p.m. Permissible shear stress is 70 MPa. Find the suitable diameter of the shaft. The maximum torque (4)

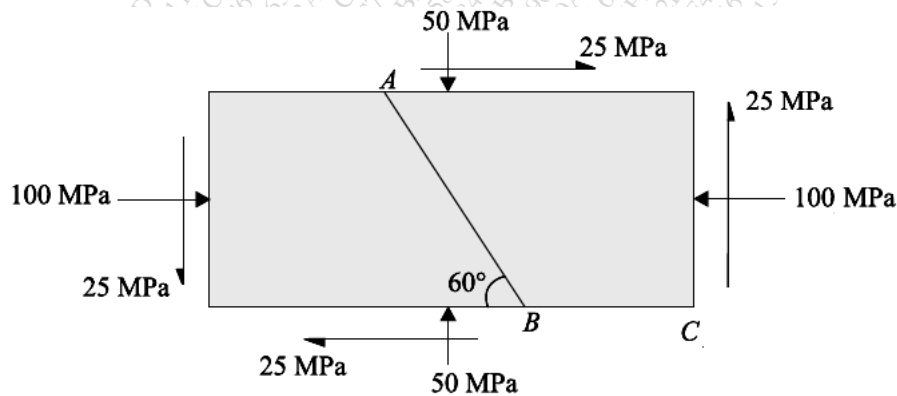
transmitted in each revolution exceeds the mean by 20%.

- b) Write the assumptions made in the theory of simple bending. (4)
- c) A cantilever beam is rectangular in section, width = 80 mm, depth = 120 mm. (4)
If cantilever is subjected to a point load of 6 kN at free end and the bending stress is not to exceed 40 MPa, find the span of the cantilever beam.

- Q. 4** a) A square section 450 mm x 450 mm is subjected to point load of 900 kN at an eccentricity of 75 mm along one of its axis of cross section. Find the stresses at four corners. Find the maximum uniaxial eccentricity, if permissible tensile stress in masonry is 2.0 N/mm^2 . (4)

OR

- a) Write the assumptions made in deriving equation of Crippling load using Euler's theory of long columns. (4)
- b) Obtain an expression for Euler's critical load for a column hinged at both ends (8)
- Q. 5** a) A point in a strained material is subjected to the stresses as shown in the figure. Find the normal and shearing stresses on the section AB inclined at an angle of 60° with x-x axis. Also find the resultant stress on the section. (6)



- b) An element in a strained body is subjected to a compressive stress of 200 MPa and a clockwise shear stress of 50 MPa on the same plane. Calculate the values of normal and shear stresses on a plane inclined at 35° with the compressive stress. Also calculate the value of maximum shear stress. (6)
- Q. 6** a) Explain: The maximum Principal Stress theory of failure. (4)
- b) A circular bar is subjected to a tensile force of 20 kN along with a transverse shear force of 10 kN. Determine the diameter of bar using Maximum Principal Stress, Maximum Principal Strain, and Maximum Shear Stress failure theory. Take: Yield strength = 250 MPa, factor of safety = 2, and Poisson's ratio = 0.3 (8)

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